

1.Created VPC with CIDR 192.168.1.0/24

VPC dashboard

EC2 Global View

Filter by VPC

Virtual private cloud

Subnets

Route tables

Internet gateways

Egress-only internet gateways

Carrier gateways

DHCP option sets

Elastic IPs

Managed prefix lists

NAT gateways

Peering connections

Route servers

Security

Network ACLs

vpc-0b6ab159960f815f5 / VPC1

Details

Info

VPC ID

vpc-0b6ab159960f815f5

DNS resolution

Enabled

Main network ACL

acl-03519345a4aa2a910

IPv6 CIDR (Network border group)

-

State

Available

Tenancy

default

Default VPC

No

Network Address Usage metrics

Disabled

Block Public Access

Off

DHCP option set

dopt-0c95a11ad0b18d3f8

IPv4 CIDR

192.168.1.0/24

Route 53 Resolver DNS Firewall rule groups

-

DNS hostnames

Disabled

Main route table

rtb-075688977c7250b5f

IPv6 pool

-

Owner ID

255914130942

Resource map

CIDRs

Flow logs

Tags

Integrations

IPv4 CIDRs

Info

Address family

IPv4

CIDR

192.168.1.0/24

Status

Associated

Edit CIDRs

2.Created Subnet in the VPC> Attached IGW to VPC> Created RT for Subnet for internet access.

Subnets

subnet-05f606c05c19f9a1e

Subnet1

Details

Info

Subnet ID

subnet-05f606c05c19f9a1e

IPv4 CIDR

192.168.1.0/28

Availability Zone

us-east-1c

Route table

rtb-02ce167f08123801c | RT-Subnet1

Auto-assign IPv6 address

No

IPv4 CIDR reservations

-

Resource name DNS A record

Disabled

Subnet ARN

arn:aws:ec2:us-east-1:255914130942:subnet/subnet-05f606c05c19f9a1e

Available IPv4 addresses

9

Availability Zone ID

use1-az1

Network ACL

acl-03519345a4aa2a910

Auto-assign customer-owned IPv4 address

No

IPv6 CIDR reservations

-

Resource name DNS AAAA record

Disabled

State

Available

IPv6 CIDR

-

Network border group

us-east-1

Default subnet

No

Customer-owned IPv4 pool

-

IPv6-only

No

DNS64

Disabled

Block Public Access

Off

IPv6 CIDR association ID

-

VPC

vpc-0b6ab159960f815f5 | VPC1

Auto-assign public IPv4 address

No

Outpost ID

-

Hostname type

IP name

Owner

255914130942

igw-00df5551346bbe6f7 / IGW-VPC1

Details

Info

Internet gateway ID

igw-00df5551346bbe6f7

State

Attached

VPC ID

vpc-0b6ab159960f815f5 | VPC1

Owner

255914130942

Tags

Search tags

Key

Value

Name

IGW-VPC1

Manage tags

rtb-02ce167f08123801c / RT-Subnet1

Actions

Details

Route table ID

rtb-02ce167f08123801c

VPC

vpc-0b6ab159960f815f5 | VPC1

Main

No

Owner ID

255914130942

Explicit subnet associations

subnet-05f606c05c19f9a1e / Subnet1

Edge associations

-

Routes

Subnet associations

Edge associations

Route propagation

Tags

Routes (2)

Both

Edit routes

Filter routes

< 1 >

Destination	Target	Status	Propagated
0.0.0.0/0	igw-00df5551346bbe6f7	Active	No
192.168.1.0/24	local	Active	No

3. Launched an Amazon Linux EC2 in the subnet and edited security group rules (Port 22, Port 80)

Instance summary for i-08e652118dd83bf7c (EC2)

Updated less than a minute ago

Connect

Instance state

Actions

Instance ID

i-08e652118dd83bf7c

IPv6 address

-

Hostname type

IP name: ip-192-168-1-14.ec2.internal

Answer private resource DNS name

-

Auto-assigned IP address

44.203.229.229 [Public IP]

IAM Role

-

IMDSv2

Required

Operator

-

Public IPv4 address

44.203.229.229 | open address

Instance state

Running

Private IP DNS name (IPv4 only)

ip-192-168-1-14.ec2.internal

Instance type

t2.micro

VPC ID

vpc-0b6ab159960f815f5 (VPC1)

Subnet ID

subnet-05f606c05c19f9a1e (Subnet1)

Instance ARN

arn:aws:ec2:us-east-1:255914130942:instance/i-08e652118dd83bf7c

Private IPv4 addresses

192.168.1.14
192.168.1.13

Public IPv4 DNS

-

Elastic IP addresses

-

AWS Compute Optimizer finding

Opt-in to AWS Compute Optimizer for recommendations. | Learn more

Auto Scaling Group name

-

Managed

false

4. ETH0 gets automatically created with EC2, Pvt and Public IP both are present (as it is default ENI), Pvt IP already in the range of 192.168.1.0/24 as created under same CIDR of VPC and subnet.

Network interface summary for eni-0cae2eacfe4989e7a

Delete network interface

Actions

Network interface details

Network interface ID

eni-0cae2eacfe4989e7a

Network interface status

In-use

VPC ID

vpc-0b6ab159960f815f5

Owner

255914130942

Source/dest. check

True

Name

-

Interface type

Elastic network interface

Subnet ID

subnet-05f606c05c19f9a1e

Requester ID

-

Managed

False

Description

-

Security groups

sg-017dcc85d5ae99063 (launch-wizard-1)

Availability Zone

us-east-1c

Requester-managed

False

Operator

-

Private IPv4 address

192.168.1.14

Public IPv4 address

44.203.229.229

Secondary public IPv4 addresses

-

Private IPv4 DNS

-

Public IPv4 DNS

-

Secondary private IPv4 addresses

-

Elastic Fabric Adapter

False

IPv6 addresses

-

Association ID

-

5. Created 2nd network interface ETH1 and then assigned the IP in range of 10.0.0.0/24 through EC2

Network interface summary for eni-0e71435f0ecd19b44		
▼ Network interface details		
Network interface ID eni-0e71435f0ecd19b44	Name -	Description ETH1
Network interface status In-use	Interface type Elastic network interface	Security groups sg-0abe78f333bd49d59 (default)
VPC ID vpc-0b6ab159960f815f5	Subnet ID subnet-05f606c05c19f9a1e	Availability Zone us-east-1c
Owner 255914130942	Requester ID -	Requester-managed False
Source/dest. check True	Managed False	Operator -
▼ IP addresses		
Private IPv4 address 192.168.1.13	Private IPv4 DNS -	Elastic Fabric Adapter False
Public IPv4 address -	Public IPv4 DNS -	IPv6 addresses -
Secondary public IPv4 addresses -	Secondary private IPv4 addresses -	Association ID -

-SSH into EC2

-ip addr show

>> We can see enX0 (default ENI) and enX1 (Custom ENI)

-sudo nano /etc/sysconfig/network-scripts/ifcfg-enX1

-Pasted:

DEVICE=enX1

BOOTPROTO=static

ONBOOT=yes

IPADDR=10.0.0.10 //assigned an IP to ETH1 within 10.0.0.0/24

NETMASK=255.255.255.0

- sudo ip addr add 10.0.0.10/24 dev enX1

```
GNU nano 2.9.3 /etc/sysconfig/network-scripts/ifcfg-enX1
DEVICE=enX1
BOOTPROTO=static
ONBOOT=yes
IPADDR=10.0.0.10
NETMASK=255.255.255.0
```

6. Adding static route to route traffic destined for 10.0.0.0/24 through the eth1 interface.

-sudo ip route add 10.0.0.0/24 dev enX1

7. To view configurations

-ip a

```
aws nups://us-east-1.console.aws.amazon.com/ec2-instance-connect/ssn/home?region=us-east-1&connType=standard&instanceId=i-06eb2110dd... [Alt+S] United States (N. Virginia) Deepsi

[ec2-user@ip-192-168-1-14 ~]$ sudo ip addr add 10.0.0.10/24 dev enX1
[ec2-user@ip-192-168-1-14 ~]$ sudo ip route add 10.0.0.0/24 dev enX1
RTNETLINK answers: File exists
[ec2-user@ip-192-168-1-14 ~]$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enX0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 9001 qdisc fq_codel state UP group default qlen 1000
    link/ether 02:3b:29:e2:1c:0f brd ff:ff:ff:ff:ff:ff
    altname eni-0cae2eacfe4989e7a
    altname device-number-0.0
    inet 192.168.1.13/28 metric 512 brd 192.168.1.15 scope global dynamic enX0
        valid_lft 3180sec preferred_lft 3180sec
    inet6 fe80::3b:29ff:fe2:1c0f/64 scope link proto kernel_l1
        valid_lft forever preferred_lft forever
3: enX1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 9001 qdisc fq_codel state UP group default qlen 1000
    link/ether 02:8a:b3:ee:2b:0d brd ff:ff:ff:ff:ff:ff
    altname eni-0e71435f0ecd19b44
    altname device-number-1.0
    inet 192.168.1.13/28 metric 513 brd 192.168.1.15 scope global dynamic enX1
        valid_lft 3263sec preferred_lft 3263sec
    inet 10.0.0.10/24 scope global enX1
        valid_lft forever preferred_lft forever
    inet6 fe80::8a:b3:ee:2b:0d/64 scope link proto kernel_l1
        valid_lft forever preferred_lft forever
[ec2-user@ip-192-168-1-14 ~]$
```

8. To check Connectivity

- ping 8.8.8.8

-ping 10.0.0.10

-ping <random IP>

```
link/ether 02:3b:29:e2:1c:0f brd ff:ff:ff:ff:ff:ff
altname eni-0cae2eacfe4989e7a
altname device-number-0.0
inet 192.168.1.13/28 metric 512 brd 192.168.1.15 scope global dynamic enX0
    valid_lft 3180sec preferred_lft 3180sec
inet6 fe80::3b:29ff:fe2:1c0f/64 scope link proto kernel_l1
    valid_lft forever preferred_lft forever
3: enX1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 9001 qdisc fq_codel state UP group default qlen 1000
    link/ether 02:8a:b3:ee:2b:0d brd ff:ff:ff:ff:ff:ff
    altname eni-0e71435f0ecd19b44
    altname device-number-1.0
    inet 192.168.1.13/28 metric 513 brd 192.168.1.15 scope global dynamic enX1
        valid_lft 3263sec preferred_lft 3263sec
    inet 10.0.0.10/24 scope global enX1
        valid_lft forever preferred_lft forever
    inet6 fe80::8a:b3:ee:2b:0d/64 scope link proto kernel_l1
        valid_lft forever preferred_lft forever
[ec2-user@ip-192-168-1-14 ~]$ ping 8.8.8.8
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data:
64 bytes from 8.8.8.8: icmp_seq=1 ttl=113 time=1.52 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=113 time=1.95 ms
64 bytes from 8.8.8.8: icmp_seq=3 ttl=113 time=1.46 ms
64 bytes from 8.8.8.8: icmp_seq=4 ttl=113 time=1.50 ms
64 bytes from 8.8.8.8: icmp_seq=5 ttl=113 time=1.92 ms
^C
--- 8.8.8.8 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4007ms
rtt min/avg/max/mdev = 1.463/1.670/1.949/0.217 ms
[ec2-user@ip-192-168-1-14 ~]$
```

```
aws Search [Alt+S] United States (N. Virginia) Deepsi

valid lifetime preferred lifetime
inet 10.0.0.10/24 scope global enX1
    valid lifetime preferred lifetime forever
    inet6 fe80::8a:b2ff:fe64:2b0d/64 scope link proto kernel ll
    valid lifetime preferred lifetime forever
[ec2-user@ip-192-168-1-14 ~]$ ping 8.8.8.8
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data.
64 bytes from 8.8.8.8: icmp_seq=1 ttl=113 time=1.52 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=113 time=1.95 ms
64 bytes from 8.8.8.8: icmp_seq=3 ttl=113 time=1.46 ms
64 bytes from 8.8.8.8: icmp_seq=4 ttl=113 time=1.50 ms
64 bytes from 8.8.8.8: icmp_seq=5 ttl=113 time=1.92 ms
^C
--- 8.8.8.8 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4007ms
rtt min/avg/max/mdev = 1.463/1.670/1.949/0.217 ms
[ec2-user@ip-192-168-1-14 ~]$ ping -c 4 10.0.0.X
ping: 10.0.0.X: Name or service not known
[ec2-user@ip-192-168-1-14 ~]$ ping -c 4 10.0.0.10
PING 10.0.0.10 (10.0.0.10) 56(84) bytes of data.
64 bytes from 10.0.0.10: icmp_seq=1 ttl=127 time=0.014 ms
64 bytes from 10.0.0.10: icmp_seq=2 ttl=127 time=0.028 ms
64 bytes from 10.0.0.10: icmp_seq=3 ttl=127 time=0.036 ms
64 bytes from 10.0.0.10: icmp_seq=4 ttl=127 time=0.028 ms
--- 10.0.0.10 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3130ms
rtt min/avg/max/mdev = 0.014/0.026/0.036/0.007 ms
[ec2-user@ip-192-168-1-14 ~]$
```

```
[ec2-user@ip-192-168-1-14 ~]$ ping 192.168.0.18
PING 192.168.0.18 (192.168.0.18) 56(84) bytes of data.
^C
--- 192.168.0.18 ping statistics ---
19 packets transmitted, 0 received, 100% packet loss, time 18733ms
[ec2-user@ip-192-168-1-14 ~]$
```

i-08e652118dd83bf7c (EC2)

9. Install Nginx

- sudo yum install nginx -y
- sudo systemctl enable nginx
- sudo systemctl start nginx

```
aws Search [Alt+S] United States (N. Virginia) Deepsi

Running transaction test
transaction test succeeded.
Running transaction
Preparing : 1/1
Running scriptlet: nginx-filessystem-1:1.26.3-1.amzn2023.0.1.noarch 1/7
Installing : nginx-filessystem-1:1.26.3-1.amzn2023.0.1.noarch 1/7
Installing : nginx-mimetypes-2.1.49-3.amzn2023.0.3.noarch 2/7
Installing : libunwind-1.4.0-5.amzn2023.0.2.x86_64 3/7
Installing : gperftools-libs-2.9.1-1.amzn2023.0.3.x86_64 4/7
Installing : nginx-core-1:1.26.3-1.amzn2023.0.1.x86_64 5/7
Installing : generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch 6/7
Installing : nginx-1:1.26.3-1.amzn2023.0.1.x86_64 7/7
Running scriptlet: nginx-1:1.26.3-1.amzn2023.0.1.x86_64 7/7
Verifying : generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch 1/7
Verifying : gperftools-libs-2.9.1-1.amzn2023.0.3.x86_64 2/7
Verifying : libunwind-1.4.0-5.amzn2023.0.2.x86_64 3/7
Verifying : nginx-1:1.26.3-1.amzn2023.0.1.x86_64 4/7
Verifying : nginx-core-1:1.26.3-1.amzn2023.0.1.x86_64 5/7
Verifying : nginx-filessystem-1:1.26.3-1.amzn2023.0.1.noarch 6/7
Verifying : nginx-mimetypes-2.1.49-3.amzn2023.0.3.noarch 7/7

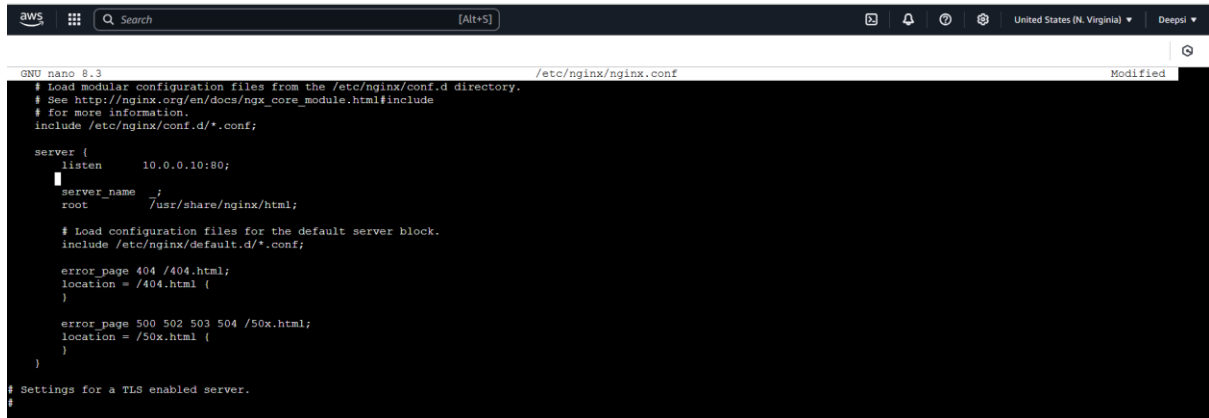
Installed:
generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch      gperftools-libs-2.9.1-1.amzn2023.0.3.x86_64      libunwind-1.4.0-5.amzn2023.0.2.x86_64
nginx-1:1.26.3-1.amzn2023.0.1.x86_64                nginx-core-1:1.26.3-1.amzn2023.0.1.x86_64      nginx-filessystem-1:1.26.3-1.amzn2023.0.1.noarch
nginx-mimetypes-2.1.49-3.amzn2023.0.3.noarch

Complete!
Created symlink /etc/systemd/system/multi-user.target.wants/nginx.service -> /usr/lib/systemd/system/nginx.service.
[ec2-user@ip-192-168-1-14 ~]$
```

10.Restrict Nginx to ETH1 only

-sudo nano /etc/nginx/nginx.conf

-Change listen to 10.0.0.10:80



```
GNU nano 2.9.3 /etc/nginx/nginx.conf Modified
# Load modular configuration files from the /etc/nginx/conf.d directory.
# See http://nginx.org/en/docs/nginx_core_module.html#include
# for more information.
include /etc/nginx/conf.d/*.conf;

server {
    listen 10.0.0.10:80;
    server_name _;
    root /usr/share/nginx/html;

    # Load configuration files for the default server block.
    include /etc/nginx/default.d/*.conf;

    error_page 404 /404.html;
    location = /404.html {
    }

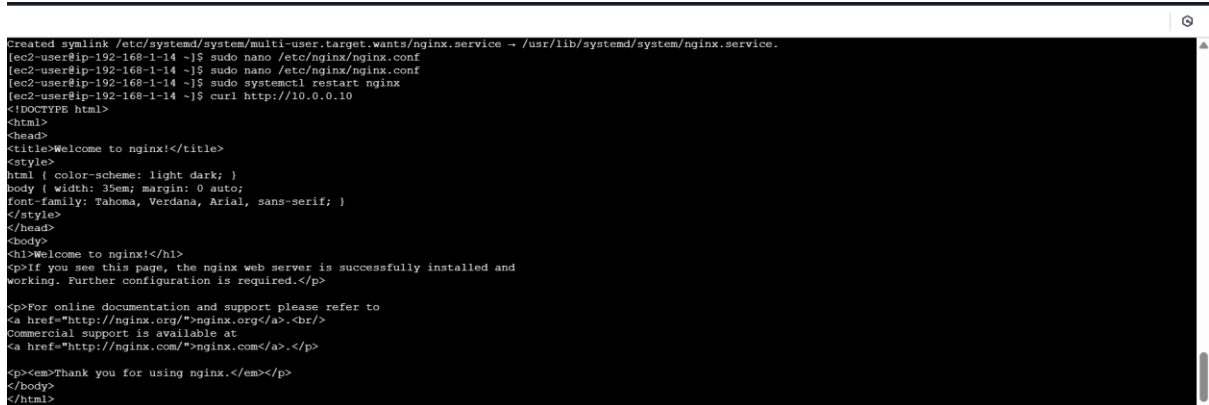
    error_page 500 502 503 504 /50x.html;
    location = /50x.html {
    }
}

# Settings for a TLS enabled server.
#
```

11. Accessing web server

- curl <http://10.0.0.10> >>gets nginx o/p

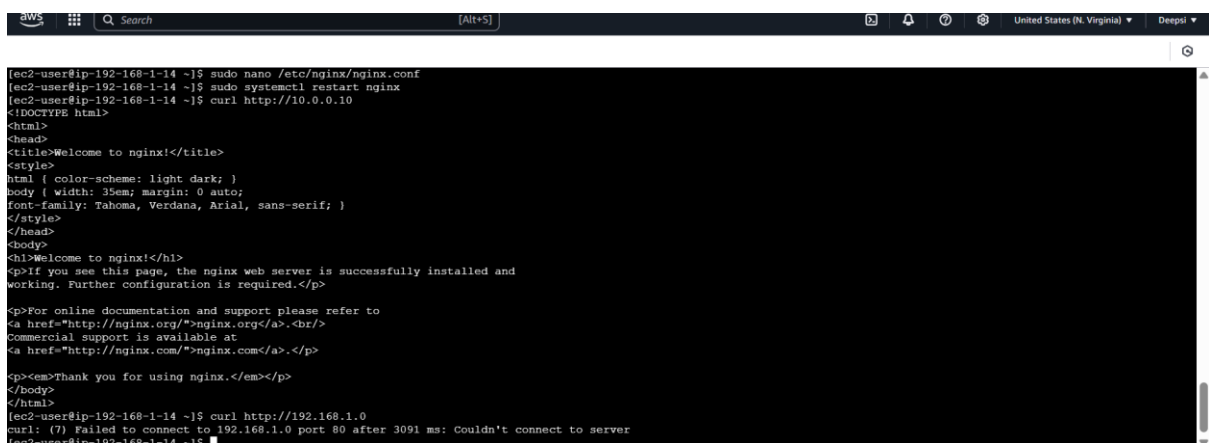
-curl <random IP> >> failed to connect message



```
Created symlink /etc/systemd/system/multi-user.target.wants/nginx.service -> /usr/lib/systemd/system/nginx.service.
[ec2-user@ip-192-168-1-14 ~]$ sudo nano /etc/nginx/nginx.conf
[ec2-user@ip-192-168-1-14 ~]$ sudo nano /etc/nginx/nginx.conf
[ec2-user@ip-192-168-1-14 ~]$ sudo systemctl restart nginx
[ec2-user@ip-192-168-1-14 ~]$ curl http://10.0.0.10
<!DOCTYPE html>
<html>
<head>
<title>Welcome to nginx!</title>
<style>
html { color-scheme: light dark; }
body { width: 35em; margin: 0 auto;
font-family: Tahoma, Verdana, Arial, sans-serif; }
</style>
</head>
<body>
<h1>Welcome to nginx!</h1>
<p>If you see this page, the nginx web server is successfully installed and
working. Further configuration is required.</p>

<p>For online documentation and support please refer to
<a href="http://nginx.org/">http://nginx.org/</a>.<br/>
Commercial support is available at
<a href="http://nginx.com/">http://nginx.com/</a>.</p>

<p><em>Thank you for using nginx.</em></p>
</body>
</html>
```



```
GNU nano 2.9.3 /etc/nginx/nginx.conf Modified
# Load modular configuration files from the /etc/nginx/conf.d directory.
# See http://nginx.org/en/docs/nginx_core_module.html#include
# for more information.
include /etc/nginx/conf.d/*.conf;

server {
    listen 10.0.0.10:80;
    server_name _;
    root /usr/share/nginx/html;

    # Load configuration files for the default server block.
    include /etc/nginx/default.d/*.conf;

    error_page 404 /404.html;
    location = /404.html {
    }

    error_page 500 502 503 504 /50x.html;
    location = /50x.html {
    }
}

# Settings for a TLS enabled server.
#
```

